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The herbal formula CGX ameliorates the expression of vascular endothelial growth factor in alcoholic liver fibrosis

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ABSTRACT

Ethnopharmacologic relevance: The Chunggan extract (CGX) is a traditional herbal formula prescribed for patients suffering from various liver diseases, including alcoholic liver disease, in which the mechanism of CGX action remains unclear. This study aimed to investigate the anti-hepatic fibrosis effects of CGX and its underlying mechanisms in alcohol-induced rat livers.

Materials and methods: To elucidate the mechanism of action of CGX, we evaluated gene expression profiles in the livers of rats treated with 30% alcohol and anti-fibrotic doses of CGX of 100 and 200 mg/kg/day at 1 day, and 2 and 4 weeks using microarrays. The mRNA and protein expression levels of vascular endothelial growth factor (VEGF), one of the candidate genes selected in this study, in alcohol-induced rat livers were measured by real-time PCR and enzyme-linked immunosorbent assays, respectively.

Results: We identified 4128 genes as differentially expressed by at least twofold between alcohol-only– and alcohol–CGX–fed rats at various doses and time points, compared to naïve control animals. Twenty-three of these genes were associated with liver fibrosis and oxidative stress based on the GeneCards database, resulting in $p < 0.05$ by ANOVA between the alcohol-only and alcohol–CGX groups. Especially, Vegf was decreased in CGX 200 mg/kg/day-fed rat livers at all time points evaluated, and mRNA and protein levels at the 4-week time point were validated.

Conclusion: These gene expression profiles provide a better understanding of the mechanisms underlying the anti-fibrotic effects of CGX. Suppression of VEGF may play a critical role in anti-fibrotic action of CGX in alcoholic liver injury.

1. Introduction

Prolonged alcohol consumption is the leading cause of chronic liver disease worldwide (Williams, 2006). It has been reported that alcohol intake of 60–80 g/day or 75–100 mL/day in men and 20 g/day or 25 mL/day in women for 20 years or more significantly increases the risk of hepatitis and fibrosis by 7–47% (Mandayam et al., 2004). Despite substantial progress in the last several decades, the mechanisms involved in liver fibrosis are not yet fully understood (Bataller and Brenner, 2005; Friedman et al., 2007). It is known that during the onset of liver fibrosis, hepatic stellate cells (HSCs) are activated through autocrine or paracrine stimulation by cytokines such as transforming growth factor-β (TGF-β), platelet-derived growth factor (PDGF), connective tissue growth factor (CTGF), leptin, and vascular endothelial growth factor (VEGF) (Hamada et al., 2005; Hetzel et al., 2005; Chaudhary et al., 2007; Lee and Friedman, 2011). In experimental studies, the VEGF expression has been noted to significantly increase during liver fibrosis development, and in mouse models, VEGF has been observed to participate in hepatic fibrogenesis (Corpechot et al., 2002; Yoshiji et al., 2003).

In recent years, there has been an increasing interest in the hepatoprotective and anti-hepatofibrotic effects of herbal medicines (Rino et al., 2006; Kwak et al., 2011). The Chunggan extract (CGX) is a commercially available variant of a traditional herbal medicine, consisting of 13 herbs involved in “cleaning the liver.” This herbal medicine had been originally described in an ancient medical book, *Donguibogam*, which was designated as a UNESCO documentary heritage in 2009 (UNESCO, 2009). Since 2001, CGX has been used as a remedy for patients with chronic liver disorders related to alcohol consumption in Korea (Cho et al., 2000). In a previous study, the hepatoprotective and anti-hepatofibrotic properties of CGX have

Abbreviations: Adh, alcohol dehydrogenase; Aldh, aldehyde dehydrogenase; CGX, Chunggan extract; CTGF, connective tissue growth factor; HPLC, high-performance liquid chromate graphy; HSC, hepatic stellate cells; PDGF, platelet-derived growth factor; PPAR, peroxisome proliferator-activated receptor; p53, protein 53; TGF, transforming growth factor; VEGF, vascular endothelial growth factor

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been demonstrated in a carbon tetrachloride (CCl_4)-induced acute injury model (Hu et al., 2008). Furthermore, the anti-fibrotic effects of CGX have also been reported using a 4-week dimethylnitrosamine (DMN) model (Shin et al., 2006; Wang et al., 2010) and a 12-week thioacetamide (TAA)-induced chronic liver injury model (Kwak et al., 2011). However, to the best of our knowledge, the molecular mechanisms underlying the effect of CGX on alcoholic liver fibrosis have not yet been determined. In the present study, we have used microarrays to evaluate the changes in gene expression in a rat model of alcohol-induced hepatic fibrosis, and have examined the anti-fibrotic effects of CGX treatment.

2. Materials and methods

2.1. Preparation and reproducibility of CGX

CGX consists of 13 herbs, including 5 g each of Artemisia Capillaris Herba, Trionycis Carapax, and Raphani Semen; 3 g each of Atractylodis Rhizoma Alba, Hoelen, Alismatis Rhizoma, Atractylodis Rhizoma, and Salviae Miltiorrhizae Radix; 2 g each of Polyporus, Aurantii Immature Fructus, and Amomi Fructus; and 1 g each of Glycyrrhizae Radix and Aucklandiae Radix (Table 1). All the herbs were verified by a herbology professor from Oriental Medicine Collage, Daejeon University, and were in compliance with the Korean Pharmacopoeia standards. CGX is manufactured by Kyoungbang Pharmacy (Incheon, Korea) following the approved Good Manufacturing Practice (GMP) of the Korea Food and Drug Administration (KFDA), according to over-the-counter Korean monographs. Briefly, 120 kg of the 13-herb mixture was boiled in 1200 L of distilled water for 4 h at 100 °C, filtered using a 300-mesh filter (50 μm), condensed, and then lyophilized. The CGX extract obtained satisfied the herb, heavy metals, general bacteria, fungi, and specific pathogens criteria, which was determined by carrying out the respective confirmation tests, and the final yield from the original dried mixture was 10.1% (w/w). The lyophilized CGX extract (100 mg) was dissolved in 50% methanol (20 mL), mixed well, and the solution was filtered through an Acrodisc® LC 13 mm Syringe filter (0.45 μm pore size, Ann Arbor, MI).

2.2. Quantitative analysis of CGX using high-performance liquid chromatography

The reference standard stock solutions of scopoletin, liquiritin, naringin, esculetin, rosmarinic acid, salvianolic acid B, poncirin, glycyrrhizin, and tanshinone IIA (200 μg/mL) were prepared in methanol and stored at < 4 °C. The standard solutions were prepared using six concentrations of diluted solutions (methanol). All the calibration curves were generated by assessing the peak areas for the six concentrations in the range of 3.13–100 μg/mL for all standard samples. The linearity of the peak area (y) vs. concentration (x, μg/mL) curve for each component was used to calculate the contents of the main components in CGX. Quantitative analysis was performed under concurrent conditions using an 1100 series high-performance liquid chromatography (HPLC; Agilent Technologies, Santa Clara, CA) equipped with an autosampler (G11313A), column oven (GA1316A), binary pump (G1312), diode-array-detector (DAD), and degasser (GA1379A). The analytical column comprised a Gemini C18 (4.6×250 mm^2; particle size, 5 μm; Phenomenex, Torrance, CA) and was maintained at 30 °C. The data were acquired and processed by chemstation software (Agilent Technologies, Santa Clara, CA). The mobile phase comprised 10% acetonitrile in water with 0.05% formic acid (A) and 90% acetonitrile in water (B). The gradient flow was as follows: 0–30 min, 0–20% B; 30–50 min, 20–75% B; 50–60 min, 75–100% B. The analysis was carried out at a flow rate of 1.0 mL/min, and detection was performed at a wavelength of 280 nm. The injection volume was 10 μL.

2.3. Animals and experimental design

A total of 108 specific pathogen-free male Sprague-Dawley (SD) rats (6 weeks old, 190–210 g) were purchased from Koatech (Gyeonggi-do, Korea). Prior to treatment, the animals were acclimated for 7 days to an environmentally controlled room at 22 ± 2 °C under a 12/12-h light/dark cycle, and were provided with pellet food (Koatech) and tap water *ad libitum*. The rats were randomly divided into five groups of six to nine animals each: Naïve animals received distilled water (control; $n=6$ for 1 day, 2 weeks, and 4 weeks); alcohol-only with distilled water (alcohol-only; $n=6$ for 1 day, $n=9$ for 2 and 4 weeks); alcohol + 100 or 200 mg/kg of CGX (alcohol–CGX; $n=6$ for 1 day, $n=9$ for 2 and 4 weeks); or 200 mg/kg of CGX only (CGX-only; $n=6$ for 1 day, 2 weeks, and 4 weeks). Alcohol was administered orally as 30% ethanol solution (10 mL/kg) for 1 day, 2 weeks, and 4 weeks (six times per week), except for the control and CGX-only groups. All the animals administered with 30% alcohol were also treated with distilled water or CGX (100 or 200 mg/kg) by gastric gavage 6 h before alcohol consumption.

The body weight of the rats was recorded twice per week, and the animals were euthanized under ether anesthesia at the end of the experiment. The liver tissues were removed and weighed, and stored in RNAlater (Ambion, Austin, TX) for RNA isolation or at −70 °C for protein expression analysis. The experiments were designed and performed in strict accordance with the *Guide for the Care and Use of Laboratory Animals* (NIH publication no. 85–23, revised 1985), and were approved by the Institutional Animal Care and Use Committee of Daejeon University (Animal ethical clearance number: DJUARB 2011-034 to 6).

Table 1
The herbal prescription of CGX.

Herbal Name	Scientific Name	Local Name	Place of Origin	Relative Amounts (g)
Artemisiae Capillaris Herba	*Artemisia capillaries* Thunberg	Yin Chen Hao	China (Hubei Sheng)	5
Trionycis Carapax	*Trionyx sinensis* Wiegmann	Bie Jia	China (Hubei Sheng)	5
Raphani Semen	*Raphanus sativus* Linne	Luo Fu Zi	Republic of Korea (Jecheon)	5
Atractylodis Rhizoma Alba	*Atractylodes macrocephala* Koidz	Bai Zhu	Republic of Korea (Bonghwa)	3
Hoelen	*Poria cocos* Wolf	Fu Ling	Republic of Korea (Bonghwa)	3
Alismatis Rhizoma	*Alisma orientalis* (*Sam.*) Juzepczuk	Ze Xie	Republic of Korea (Hwasun)	3
Atractylodis Rhizoma	*Atractylodes chinensis* Koidzumi	Cang Zhu	Republic of Korea (Bonghwa)	3
Salviae Miltiorrhizae Radix	*Salvia miltiorrhiza* Bunge	Dan Shen	China (Hubei)	3
Polyporus	*Polyporus umbellatus* Fries	Zhu Ling	China (Yunnan Sheng)	2
Aurantii Immaturus Fructus	*Poncirus trifoliate* Rafin	Zhi Shi	Republic of Korea (Yeongcheon)	2
Amomi Fructus	*Amomum villosum* Lour	Sha Ren	China (Yunnan Sheng)	2
Glycyrrhizae Radix	*Glycyrrhiza uralensis* Fisch.	Gan Cao	China (Nei meng gu)	1
Aucklandiae Radix	*Aucklandia lappa* Decne.	Mu Xiang	China (Yunnan Sheng)	1

2.4. Histopathological findings

For evaluating the accumulation of collagen in the liver tissue, a portion of liver tissue in 10% formalin solution was re-fixed in Bouin's solution. The paraffin-embedded liver tissue was sectioned (4-μm thickness), and Masson's trichome staining was performed. Characteristic histopathological features, such as necrosis or inflammatory cell infiltration and fibrosis, were examined under the microscope.

2.5. Determination of hydroxyproline in liver tissues

Hydroxyproline was determined by using the method described previously with slight modification (Fujita et al., 2003). Briefly, the liver tissues (200 mg) stored at −70 °C were homogenized in 2 mL of 6 N HCl and incubated overnight at 110 °C. After filtering the acid hydrolysates through a filter paper (Toyo Roshi Kaisha, Tokyo, Japan), 50 μL of the samples or hydroxyproline standards in 6 N HCl were air-dried. The dried samples were dissolved in methanol (50 μL), and subsequently, 1.2 mL of 50% isopropanol and 200 μL of chloramine-T solution were added to each sample, followed by incubation at room temperature for 10 min. Then, Ehrlich's solution (1.3 mL) was added, and the samples were incubated at 50 °C for 90 min. The optical density of the reaction product was read at 558 nm using a spectrophotometer (Cary 50; Varian, Palo Alto, CA). A standard curve was constructed using serial dilutions of 1.0 mg/mL of hydroxyproline solutions.

2.6. RNA isolation and quality control

The total RNA was isolated from the livers of each animal in the control and experimental groups using TRIzol reagent (Invitrogen, Carlsbad, CA, USA). The RNA was treated with RNase-free DNase I (Promega, Madison, WI, USA) to eliminate DNA contamination. The RNA concentration and quality were determined using a NanoDrop® ND-1000 spectrophotometer (NanoDrop Technologies Inc., Wilmington, DE, USA), and the Agilent RNA Integrity number (RIN) (Schroeder et al., 2006) was evaluated using the Agilent 2100 Bioanalyzer and the Agilent RNA 6000 Nano kit (Agilent Technology, Santa Clara, CA, USA). For all the samples, the absorbance ratio at 260:280 nm was >1.8 and the RIN value was >8. The integrity of the RNA samples was also ascertained by the presence of distinct 28S and 18S ribosomal RNA bands in the agarose gels after electrophoretic resolution.

2.7. Microarray analysis

For control and test RNAs, target complementary RNA (cRNA) was synthesized and hybridization was performed using Agilent's Low RNA Input Linear Amplification Kit PLUS (Agilent Technology), according to the manufacturer's instructions. Briefly, 0.5 μg of total RNA from each sample was mixed with the diluted Spike mix and T7 promoter primer mix, and incubated at 65 °C for 10 min. The cDNA master mix (5 × First-strand buffer, 0.1 M dithiothreitol [DTT], 10 mM deoxynucleotide triphosphate [dNTP], RNase-Out, and Moloney Murine Leukemia Virus Reverse Transcriptase [MMLV-RT]) was prepared and added to the reaction mix. The samples were incubated at 40 °C for 2 h and dsDNA synthesis was terminated by incubation at 65 °C for 15 min. The transcription master mix was prepared according to the manufacturer's protocol (4 × Transcription buffer, 0.1 M DTT, NTP mix, 50% polyethyleneglycol, RNase-Out, inorganic pyrophosphatase, T7-RNA polymerase, and Cyanine [Cy] 3/5-CTP). Transcription was performed by adding the transcription master mix to the dsDNA reaction samples and incubating at 40 °C for 2 h. The amplified and labeled cRNA was purified on RNase Mini Spin Columns (Qiagen, Hilden, Germany), according to the manufacturer's instruction. The labeled cRNA target was quantified using the NanoDrop® ND-1000 spectrophotometer. After checking the labeling efficiency, 750 ng of each Cy3-labeled and Cy5-labeled cRNA target were mixed and fragmented by adding 10 × blocking agent and 25 × fragmentation buffer, followed by incubation at 60 °C for 30 min. The fragmented cRNA was resuspended with 2 × hybridization buffer and directly added to an assembled Agilent Whole Rat Genome Oligo Microarray (Rat GE 4 × 44K v3). The arrays were hybridized at 65 °C for 17 h using an Agilent Hybridization Oven, and the hybridized microarrays were washed according to the manufacturer's protocol.

2.8. Data acquisition and analysis

The hybridization images were analyzed using an Agilent DNA microarray scanner, and data quantification was performed using Agilent Feature Extraction software. The average fluorescence intensity of each spot was calculated and the local background was subtracted. All data normalization and fold-change calculations were performed using GeneSpring GX 7.3 (Agilent Technology). The genes were filtered by removing flag-out genes in each experiment. Intensity-dependent normalization was carried out using locally weighted scatterplot smoothing (LOWESS), where the ratio was reduced to the residual of the LOWESS fit of the intensity vs. ratio curve. The average normalized ratios were calculated by dividing the average normalized signal channel intensity by the average normalized control channel intensity. Functional annotation of genes was performed according to the Gene Ontology™ Consortium (http://www.geneontology.org/index.shtml) using GeneSpring GX 7.3. Gene classification was based on searches using GeneCards (http://www.genecards.org/), BioCarta (http://www.biocarta.com/), DAVID (http://david.abcc.ncifcrf.gov/), and Medline databases (http://www.ncbi.nlm.nih.gov/).

2.9. Real-time PCR

The cDNA was synthesized from 1 μg of total RNA according to the manufacturer's instructions (MMLV Reverse Transcriptase; Invitrogen, Carlsbad, CA, USA). The reactions were carried out on an ABI Prism® 7900HT Sequence Detection System (PE Applied Biosystems, Foster City, CA, USA) and relative transcript levels were determined using β-actin as the endogenous control. Each reaction was performed in triplicate. The thermal cycling conditions were 95 °C for 10 min, followed by 45 cycles of 95 °C for 10 s, 60 °C for 15 s, and 72 °C for 20 s. The data were analyzed using the Sequence Detection System software (SDS version 2.0, PE Applied Biosystems). PCR was performed for nine genes, including β-actin. The sequences of the forward and reverse primers were as follows: *Adh4* (NM_017270), (forward) 5′-CCC AAA CCT GGT TAC TGA CTA CAA-3′ and (reverse) 5′-GGT CAA TCG CGT CGT TGA TT-3′; *Adh6* (NM_001012084), (forward) 5′-GCC ATC AAC TCT GCC AAG GT-3′ and (reverse) 5′-GCT GCT CCT GCT GCT TTA CA-3′; *Aldh1a1* (NM_022407), (forward) 5′-GAG TCT TCT ACC ATC AAG GCC AAT-3′ and (reverse) 5′-GCT CGC TCA ACA CTC TTT CTC A-3′; *Aldh1a7* (NM_017272), (forward) 5′-TTG GAT AGT GCT GTT GAG TTT GC-3′ and (reverse) 5′-CGT AAA TGG ACT CCT CAA CAA AAA-3′; *Aldh1b1* (NM_001011975), (forward) 5′-TGG ACG GCG AAC ATT TCT G-3′ and (reverse) 5′-AGC TTC CAA CCC TGC ATG AC-3′; *Vegfa* (NM_001110334), (forward) 5′-GAC GGG CCT CTG AAA CCA T-3′ and (reverse) 5′-TGC AGC CTG GGA CCA CTT-3′; *Pdgf-β* (NM_031524), (forward) 5′-ACC ACT CCA TCC GCT CCT TT-3′ and (reverse) 5′-TGT GCT CGG GTC ATG TTC AA-3′; *Tgf-β1* (NM_021578), (forward) 5′-AGG AGA CGG AAT ACA GGG CTT T-3′ and (reverse) 5′-AGC AGG AAG GGT CGG TTC AT-3′; *Ctgf* (NM_022266), (forward) 5′-GTG TGT GAT GAG CCC AAG GA-3′ and (reverse) 5′-CAG TTG GCT CGC ATC ATA GTT G-3′;

and β-actin (NM_031144), (forward) 5′-CTA AGG CCA ACC GTG AAA AGA T-3′ and (reverse) 5′-GAC CAG AGG CAT ACA GGG ACA A-3′.

2.10. VEGF quantification

The VEGF levels in hepatic tissue were quantified using an enzyme-linked immunosorbent assay (ELISA) kit (rat VEGF, R&D Systems, Minneapolis, MN, USA), according to the manufacturer's instructions. Briefly, the central liver lobe was harvested, snap-frozen in liquid nitrogen, and pulverized using a pre-chilled mortar and pestle on dry ice. Then, approximately 100 mg of each liver was homogenized in 1.5 mL of NP-40 lysis buffer (50 mM HEPES, pH 7.5; 150 mM NaCl, 10% glycerol, 1.5 mM $MgCl_2$, 1 mM EDTA, 100 mM NaF, and 1% NP-40) containing protease inhibitors (10 μg/mL of aprotinin, 10 μg/mL of leupeptin, and 1 mM 4-[2-aminoethyl]-benzenesulfonyl-fluoride, hydrochloride). The liver homogenates were centrifuged at 14,000 rpm for 15 min. The surface lipid layer was removed, and the supernatant was stored at −80 °C for ELISA. Prior to analysis, the supernatant was transferred to a new 1.5-mL microfuge tube and the VEGF expression was determined by measuring the absorbance at 492 nm. The concentrations were calculated using a standard curve generated with specific standards provided by the manufacturer.

Table 2
Quantitative analysis of CGX and its reference compounds.

Compound	Retention time (min)	Mean (μg/g)	SD
Scopoletin	22.13	5.06	0.14
Liquritin	23.26	17.98	0.12
Naringin	28.30	228.79	6.50
Esculetin	31.21	ND[a]	ND
Rosmarinic acid	32.05	7.15	0.08
Salvianolic acid B	37.63	21.86	0.31
Poncirin	38.22	6.63	0.10
Glycyrrhizic acid	46.38	51.45	0.86
Tanshinone IIA	59.12	1.82	0.13

[a] ND, not detected.

2.11. Statistical analysis

The results obtained are expressed as the means ± standard deviation (SD). The statistical significance of differences among the groups was analyzed by one-way analysis of variance (ANOVA), followed by Student's unpaired *t*-test. In all the analyses, *p* values of <0.05 or <0.01 indicated statistical significance.

3. Results

3.1. Fingerprint analysis of CGX

The standard curves for the nine components containing scopoletin, liquiritin, naringin, esculetin, rosmarinic acid, Salvianolic acid B, poncirin, glycyrrhizin, and tanshinone IIA were $y=11.887x-8.247$ ($R^2=0.999$), $y=16.447x-11.451$ ($R^2=0.999$), $y=15.211x-11.069$ ($R^2=0.999$), $y=11.713x-9.079$ ($R^2=0.999$), $y=17.497x-16.149$ ($R^2=0.999$), $y=7.937x-16.028$ ($R^2=0.998$), $y=17.155x-8.843$ ($R^2=0.999$), $y=0.728x-0.245$ ($R^2=0.991$), and $y=29.448x-21.802$ ($R^2=0.999$), respectively. The components obtained from HPLC analysis of CGX and standard mixtures were detected at 280 nm. The retention times of each component were 22.13 min (scopoletin), 23.16 min (liquiritin), 28.30 min (naringin), 31.21 min (esculetin), 32.05 min (rosmarinic acid), 37.63 min (salvianolic acid B), 38.22 min (poncirin), 46.38 min (glycyrrhizin), and 59.12 min (tanshinone IIA). The contents of each component were in the range of 1.823–228.79 μg/g (Table 2).

3.2. Effect of CGX on gene expression profiles

The 4-week oral administration of 30% ethanol notably induced hepatofibrotic change, which was determined by using Masson's trichome staining (Fig. 1A); however, this change was not observed at 2 weeks (data not shown). This histopathological finding was noted to be in accordance with the accumulation of hydroxyproline content in hepatic tissues. The level of hydroxyproline content increased up to 2.4 folds following 4-week ethanol consumption, but it was not induced within 2 weeks (Fig. 1B). On the other hand, the CGX

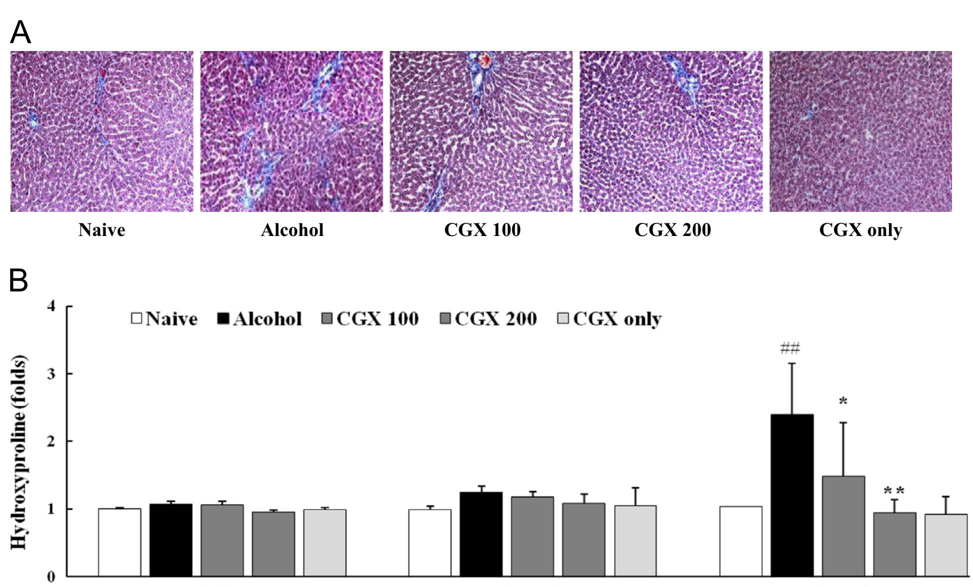

Fig. 1. Effect of alcohol and CGX on collagen deposition. At the end of the each experiment, all animals were sacrificed. The liver sections were examined with Masson's trichrome staining at 4-week under an optical microscope (×200 magnification). (A) The collagen accumulations were determined by hydroxyproline. (B) The hydroxyproline contents were normalized with 1 day of normal group and represented by fold changes. $^{\#\#}p<0.01$ compared with the naïve group; $^{**}p<0.01$ compared with the alcohol-only group.

treatment markedly reduced the histopathological change (Fig. 1A) and also significantly attenuated the alteration of hydroxyproline level ($p < 0.05$ for 100 mg/kg of CGX and $p < 0.01$ for 200 mg/kg of CGX; Fig. 1B).

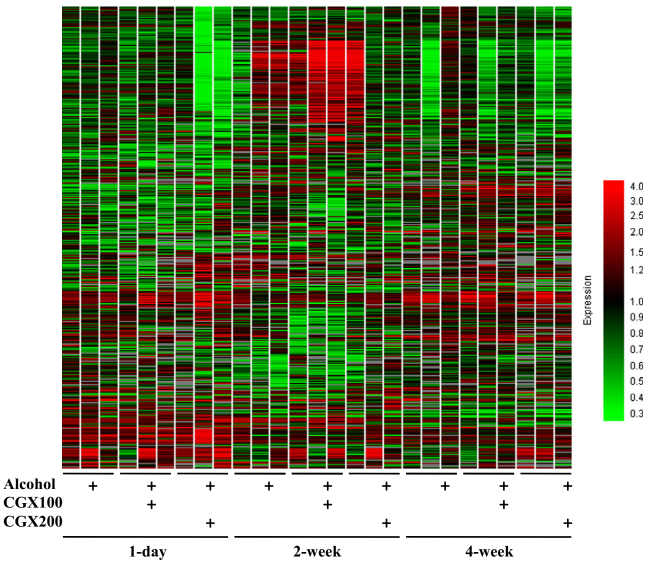

Fig. 2. Effect of alcohol and CGX on gene expression profiles in rat liver tissues. Total RNA was extracted from naïve and alcohol-only (30%) or alcohol–CGX (100 or 200 mg/kg)-treated rat liver tissues after 1 day, and 2 and 4 weeks of treatment. Alcohol-only samples were labeled with Cy3 and alcohol–CGX samples with Cy5, except for the CGX-only group. Microarray analysis was performed using biological triplicates for each group. Data normalization (LOWESS) was performed using GeneSpring GX 7.3 (see Agilent's GeneSpring GX software; http://www.genomics.agilent.com/). A gene set representing > 2-fold changes in at least one group is presented by hierarchical clustering analysis (red, > 2-fold change; green, < 2-fold change). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.3. Effect of CGX on gene expression profiles

The Agilent Rat GE 4 × 44K v3 microarray (Agilent Technologies, Santa Clara, CA) was used to identify genes involved in producing anti-oxidative and anti-fibrotic effects on alcohol-induced rat liver tissues upon CGX treatment. This array contains 30,367 unique genes, including 26,930 genes deposited in the National Center for Biotechnology Information database. We identified 4128 genes as differentially expressed in alcohol-only- and alcohol–CGX-fed rats, when compared with the control animals (Supplementary Table S1). These genes were selected based on at least twofold expression level change in at least one of the nine groups (alcohol-only and alcohol–CGX at 100 or 200 mg/kg/day; three time points of 1 day, 2 weeks, and 4 weeks). Analysis of these 4128 genes in a nine-group hierarchical gene cluster showed scattered expression characterized by several distinct patterns, including up- or downregulation at the three time points (Fig. 2).

3.4. Effect of CGX on pathway analysis

Biologically relevant pathways were constructed using the Database for Annotation, Visualization, and Integrated Discovery (DAVID) tools (http://david.abcc.ncifcrf.gov/). The gene lists corresponding to at least twofold up- and downregulation in the 200-mg/kg/day alcohol–CGX treatment group at the 4-week time point (see Supplementary Table S2) were uploaded to DAVID for Gene Ontology and pathway analyses. The upregulated genes included those involved in the cell cycle, lipid, fatty acid and steroid metabolism, and the p53 and PPAR signaling pathways. The downregulated genes included those involved in cell communication, lipid, fatty acid and steroid metabolism, intracellular signaling cascade, immunity and defense, and TGF-β signaling.

In addition, biologically relevant pathways were also constructed using the KEGG Mapper-Search & Color Pathway tool (http://www.genome.jp/kegg/tool/map_pathway2.html) with the

Table 3
Fibrosis and oxidative-stress related changes in gene expression in alcohol-treated rat livers.

Systematic ID	Alcohol (1 day)	A+CGX100 (1 day)	A+CGX200 (1 day)	Alcohol (2 wks)	A+CGX100 (2 wks)	A+CGX200 (2 wks)	Alcohol (4 wks)	A+CGX100 (4 wks)	A+CGX200 (4 wks)	GenBank Accession	Gene Symbol
A_64_P067659	1.22[a]	1.17	1.39	1.20	1.71	1.89	0.78	2.35	2.91	NM_022407	Aldh1a1
A_64_P057746	0.73	1.23	ND[b]	0.41	0.96	0.24	0.44	0.64	ND	NM_012517	Cacna1c
A_64_P109657	0.82	1.22	2.23	1.05	1.04	1.02	0.74	1.18	1.12	NM_012550	Ednra
A_44_P313825	0.89	1.15	0.49	0.89	0.99	0.92	0.99	0.96	0.70	NM_019335	Eif2ak2
A_44_P378484	0.62	0.83	0.50	0.84	1.90	1.00	0.82	0.81	0.60	NM_001159656	Gnas
A_43_P15993	0.28	0.41	1.02	0.66	0.29	2.29	0.93	1.04	1.15	NM_012711	Itgam
A_44_P400324	0.80	0.95	0.45	0.65	0.92	0.57	1.14	1.03	0.91	NM_019318	Maf
A_44_P653701	0.57	1.15	2.11	0.99	0.83	1.64	0.89	2.14	1.17	NM_001106536	Mybl2
A_64_P079553	0.70	0.75	1.16	1.16	0.42	1.39	0.98	ND	ND	NM_001100485	Myh8
A_64_P035940	0.90	0.84	0.48	0.89	1.19	0.79	1.14	0.80	0.72	NM_021742	Nr5a2
A_64_P051631	1.14	0.52	0.69	0.78	0.46	0.61	1.18	0.50	0.87	NM_017031	Pde4b
A_64_P085715	0.46	0.65	0.48	0.56	1.68	0.83	0.42	0.90	0.71	NM_001191934	Pkd2
A_44_P139291	0.69	0.64	0.37	0.74	0.68	0.58	1.11	0.88	0.71	NM_001135778	Prodh
A_64_P044315	0.90	0.85	0.49	0.68	0.71	0.64	0.98	0.84	0.65	NM_019163	Psen1
A_64_P137130	0.67	0.83	0.47	1.06	1.41	0.55	0.89	0.92	0.53	NM_053767	Ptpre
A_64_P051307	1.37	0.95	0.83	2.21	0.54	1.68	1.63	1.60	1.08	NM_053470	Runx2
A_44_P823749	0.81	0.83	1.41	1.26	0.76	1.61	1.55	1.22	2.09	NM_013125	Scn5a
A_64_P019295	0.66	0.74	0.48	0.73	0.79	0.67	0.90	0.88	0.77	NM_001164060	Shc1
A_44_P234624	1.03	0.82	0.49	0.93	1.05	0.72	1.15	0.99	0.78	NM_001107061	Smurf2
A_43_P21800	0.72	0.83	0.44	1.00	1.14	0.86	1.06	0.98	0.78	NM_001024254	Taok3
A_64_P099893	0.94	0.82	0.46	0.67	0.96	0.75	1.01	0.87	0.61	NM_001110334	Vegfa
A_64_P099891	0.83	0.78	0.45	0.84	1.08	0.74	0.97	0.86	0.61	NM_001110334	Vegfa
A_64_P071506	1.38	0.46	1.44	1.01	ND	1.25	1.10	0.84	ND	XM_001078643	Xylt1

[a] Data are expressed as the fold change in gene expression calculated as the ratio between the treatment (alcohol-only, alcohol [A]–CGX 100, or alcohol–CGX 200) and control (naïve) signal intensities.
[b] ND, not detected.

above-mentioned gene list (see Supplementary Fig. S1). The data suggested that alcohol–CGX treatment exerted effects on the cell cycle and MAPK, PPAR, TGF-β, and VEGF signaling (Supplementary Fig. S1).

3.5. Effect of CGX in fibrosis-related genes

To determine whether alcohol–CGX treatment affected anti-oxidative and anti-fibrotic gene expression, we identified candidate genes using the GeneCards database (http://www.genecards.org/). The alcohol-only and alcohol–CGX groups were evaluated independently by using one-way ANOVA (see Supplementary Fig. S2) and the results were schematically illustrated as Venn diagrams, as shown in Supplementary Fig. S2, and the gene lists are provided in Table 3. These data mining tools determined that the aldehyde dehydrogenase (ALDH) 1 family member A1 (*Aldh1a1*) and VEGF A (vegf) genes were up- and down-regulated by CGX in alcohol-induced rat liver tissues.

We investigated the role of CGX in alcohol-induced hepatic toxicity in terms of the anti-fibrotic effects of the *Adh* and *Aldh* gene families, along with four potential growth factors, including *Vegf*. These gene sets intersected with more than twofold changes in at least one of the nine groups (alcohol-only and alcohol–CGX at 100 or 200 mg/kg/day at three time points). The result of these intersections obtained by hierarchical clustering is presented in Fig. 3A. In the 4-week chronic ethanol-treatment group, *Adh4* expression was slightly induced in alcohol-only treatment group, when compared with the control rat liver tissues, whereas alcohol–CGX treatment at 100 and 200 mg/kg/day further induced *Adh4* expression in a dose-dependent manner (Fig. 3B). On the other hand, the *Aldh1a1* gene expression was reduced in alcohol-only-treated livers, when compared with that in control livers. However, treatment with alcohol–CGX at 100 or 200 mg/g/day induced *Aldh1a1* gene expression in a dose-dependent manner, when compared with alcohol-only treatment. In contrast, *Aldh1b1* gene expression exhibited the opposite trend, consistent with the microarray analysis (Fig. 3C). After 1 day of treatment, *Adh6* and *Aldh1a7* expression levels were significantly increased in the alcohol-CGX-treated livers (Fig. 3B and C). The results of the four anti-fibrotic growth factors, including *Vegf*, obtained by hierarchical clustering are presented in Fig. 4A. RT-PCR demonstrated dose-dependent decreases in *Vegf*, *Pdgf-β*, *Tgf-β1*, and *Ctgf* mRNA levels upon treatment with alcohol–CGX at the 4-week time point, when compared with those upon alcohol-only treatment, which exhibited increased mRNA levels of these genes (Fig. 4B and C). However, after 1 day of treatment, the expression levels of the four growth factors in the alcohol-only- and alcohol–CGX-treated groups were not significantly different (data not shown).

3.6. Effect of CGX on VEGF expression

VEGF is a key factor in liver fibrosis development, and changes in its levels were confirmed by ELISA (Fig. 4B). It was observed that 4-week treatment with alcohol–CGX at 100 and 200 mg/kg/day resulted in a slight decrease in VEGF expression, when compared with alcohol-only treatment.

4. Discussion

In the present study, as expected, histopathological examination and measurement of hydroxyproline content revealed that

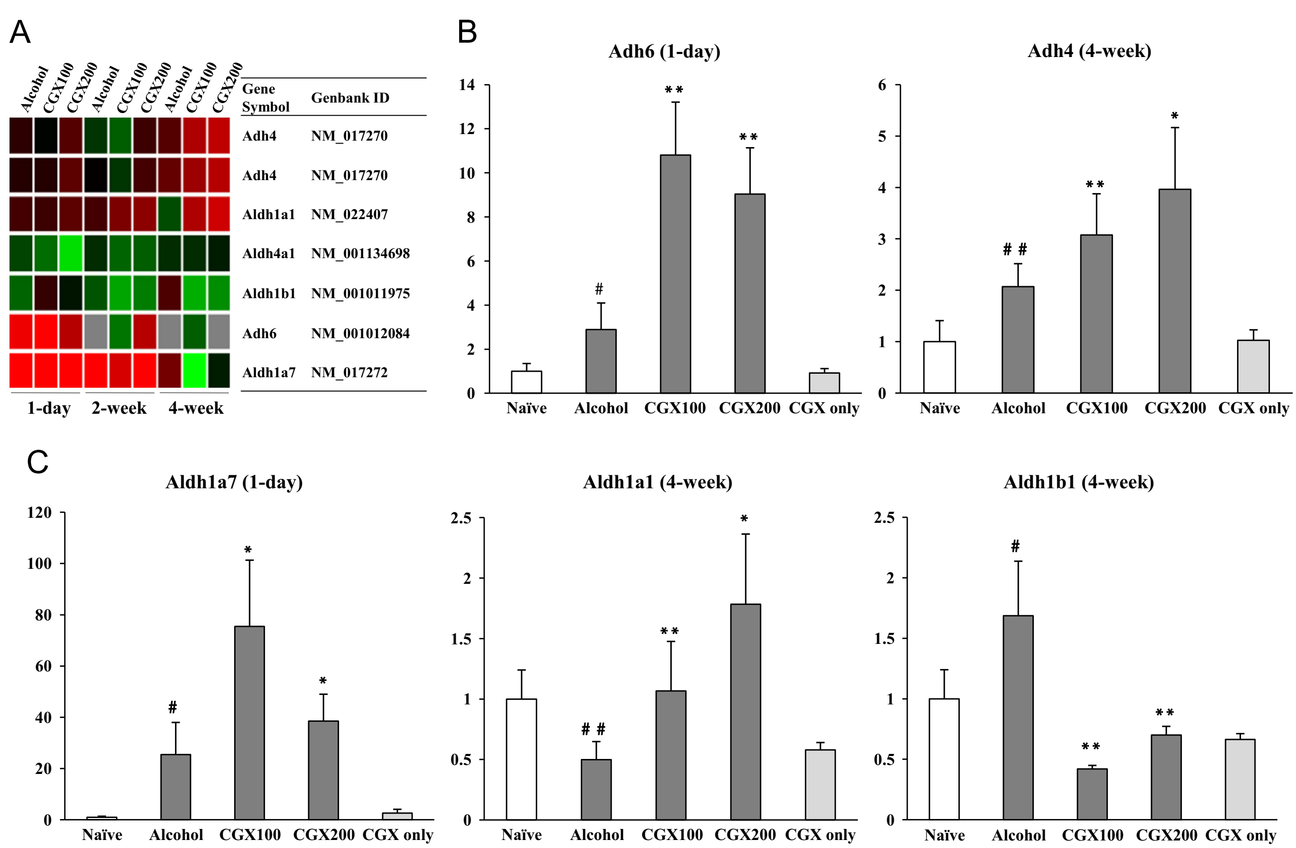

Fig. 3. Effect of CGX on *Adh* and *Aldh* mRNA expression in alcohol-induced rat liver tissues. Total RNA was extracted from naïve and alcohol-only (30%) or alcohol–CGX (100 or 200 mg/kg) or CGX-only (200 mg/kg)-treated rat liver tissues at the indicated time points (1 day or 4 weeks). (A) These three gene lists (Adh and Aldh gene families , >2-fold increase, and <2-fold decrease of gene expression in at least one group) were then individually intersected with Venn diagrams (see Supplementary Fig. S2). The result of these intersections is presented by hierarchical clustering. (B and C) Relative gene expression levels were determined by RT-PCR. Each value represents the mean ± SD of triplicate experiments. $^{\#}p < 0.05$, $^{\#\#}p < 0.01$ compared with the naïve group; $^{*}p < 0.05$, $^{**}p < 0.01$ compared with the alcohol-only group.

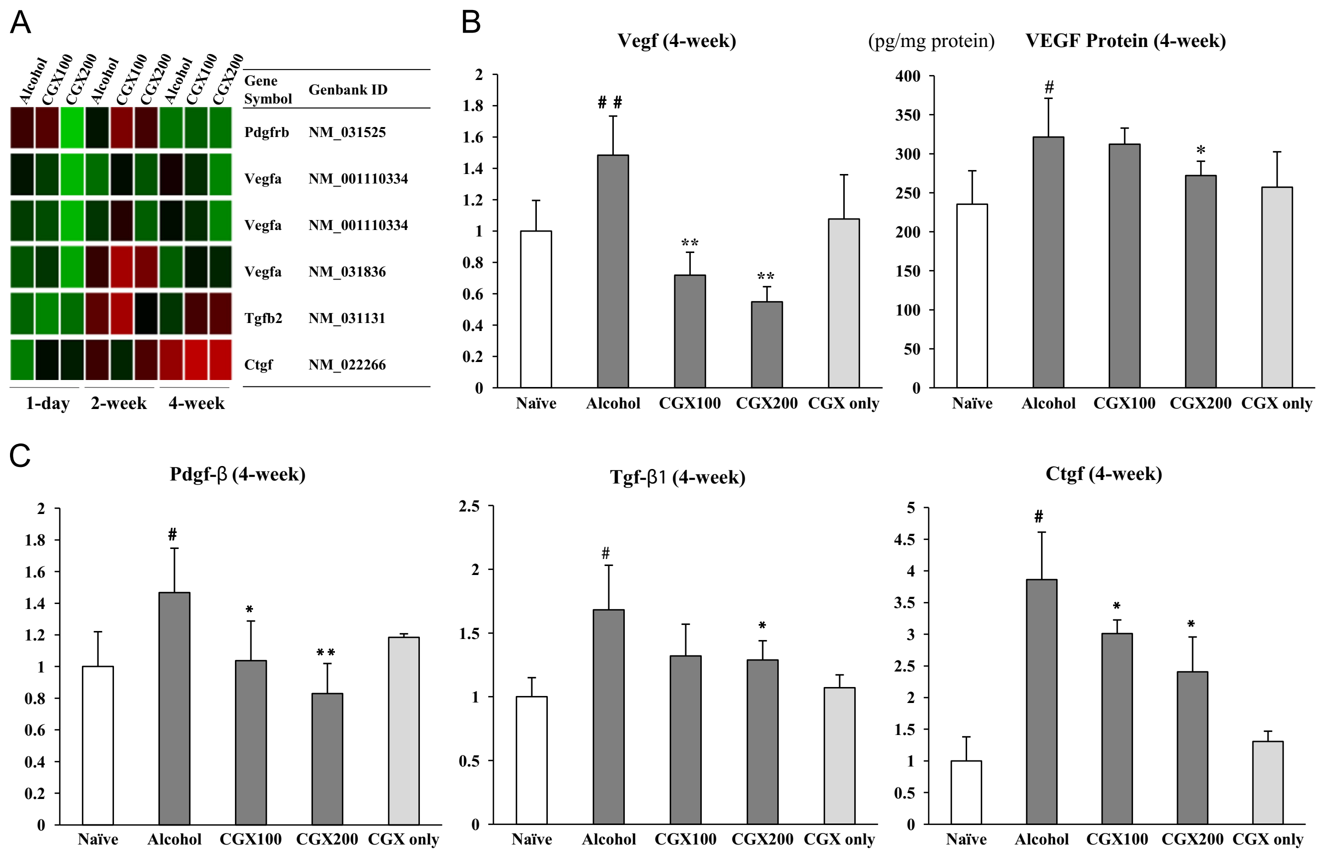

Fig. 4. Effect of CGX on *Vegf* and 3-growth factor mRNA and VEGF protein levels in alcohol-induced rat liver tissues. Total RNA and protein were extracted from naïve and alcohol-only (30%) or alcohol–CGX (100 or 200 mg/kg) or CGX-only (200 mg/kg)-treated rat liver tissues at the 4-week time point. (A) These three gene lists (potential 4 growth factors involved in the anti-fibrotic effect, >2-fold increase, and <2-fold decrease of gene expression in at least one group) were then individually intersected with Venn diagrams (see Supplementary Fig. S2). The result of these intersections is presented by hierarchical clustering. (B) *Vegf* mRNA and VEGF protein levels determined by RT-PCR (left panel) and ELISA (right panel), respectively. (C) The relative expression levels of the 3-growth factor mRNAs were determined by RT-PCR. Each value represents the mean ± SD of triplicate experiments. $^{\#}p<0.05$, $^{\#\#}p<0.01$ compared with the naïve group; $^{*}p<0.05$, $^{**}p<0.01$ compared with the alcohol-only group.

4-week oral administration of 30% ethanol notably induced hepatofibrotic changes, while CGX treatment significantly attenuated these alterations (Fig. 1A and B). It has been reported that CGX sustained anti-oxidative stress mechanisms, such as those related to glutathione, and inhibited reactive oxygen species (ROS) production in the TAA-induced chronic rat liver injury model (Kwak et al., 2011). Furthermore, CGX has been observed to exhibit anti-fibrotic effects on 4-week DMN model (Shin et al., 2006; Wang et al., 2010) as well as hepatoprotective effects on alcohol consumption model (Kim et al., 2013a, 2013b). In the present study, the underlying mechanism of the anti-fibrotic effects of CGX on alcohol-induced liver fibrosis was examined using DNA microarray analysis of gene expression.

Our results showed that VEGF is specifically modulated in alcoholic liver fibrosis following CGX treatment, indicating suppression of VEGF expression as a possible mechanism of CGX action. Alcohol-induced fibrosis is triggered by the release of fibrogenic factors such as PDGF, TGF-β, CTGF, and VEGF from activated Kupffer cells (Schuppan et al., 1995; Chaudhary et al., 2007; Gressner and Gressner, 2008; Kisseleva and Brenner, 2008; Wallace et al., 2008). However, VEGF remains the strongest angiogenic inducer (Hoeben et al., 2004) by controlling endothelial cell survival, proliferation, and postnatal angiogenesis. Furthermore, VEGF binds to the VEGFR2 receptor and mediates its biological responses through ROS (Ushio-Fukai, 2007).

In the present study, the gene expression profiles of rats treated with 30% alcohol-only and 30% alcohol–CGX at 100 or 200 mg/kg/day for 1 day, 2 weeks, and 4 weeks were compared. A total of 4128 genes were identified to be differentially expressed in the alcohol-only- and alcohol–CGX-treated rats, when compared with those in the control animals (Supplementary Table S1). As shown by the hierarchical clustering results, we expected the gene expression profiles of the alcohol-only and alcohol–CGX groups to differ, but did not expect the gene clusters to be separated in a time-dependent manner (Fig. 2). The development of tolerance through chronic ethanol intake is one of the requirements for the alcohol-research animal model (Tabakoff and Hoffman, 2000; Dyr and Taracha, 2012). Furthermore, there is a possibility for the metabolic effects of chronic ethanol intake at 2 or 4 weeks to differ, which may explain the time-dependent effects of CGX. In addition, we identified inter-individual variations in the gene expression patterns at 2 and 4 weeks of alcohol–CGX treatment (Fig. 2).

To determine the effect of CGX on chronic ethanol intake, we specifically focused on the genes altered at the 4-week time point. As shown in Supplementary Table S2 and Fig. S1, the biological effects of CGX were predicted using microarray data mining tools, such as DAVID and the KEGG Mapper. At this time point, CGX affected a variety of signal transduction pathways, including VEGF signaling. Furthermore, we also identified well-known anti-oxidative and anti-fibrotic genes affected by alcohol–CGX treatment using the GeneCards database and GeneSpring GX software by employing ANOVA, Venn diagrams, and hierarchical clustering (Table 3). Subsequently, we selected the *Adh* and *Aldh* gene families, along with four growth factors, as candidate genes affected by CGX. The expression levels of these genes in the alcohol–CGX-treated livers were confirmed by RT-PCR (Figs. 3 and 4).

Alcohol dehydrogenase (ADH) is known to oxidize ethanol to acetaldehyde and regulate ethanol-induced hepatic toxicity, although the mechanism of action remains unclear (Rikans and Moore, 1987; Zhang et al., 2005). Humans possess the following seven ADH isozymes: class I, α, β, and γ; class II, π; class III, χ; class IV, ADH7; and class V, ADH6 (http://www.genecards.org/). In rats, Adh4 and Adh6 are members of the ADH family of classes II and V, respectively. Members of this enzyme family metabolize a wide range of substrates, including ethanol, retinol, other aliphatic alcohols, hydroxysteroids, and lipid peroxidation products, and their genes are predominantly expressed in the liver (http://www.ncbi.nlm.nih.gov/). On the other hand, ALDH belongs to the ALDH family of enzymes that catalyze the chemical transformation from acetaldehyde to acetic acid. ALDH is the second enzyme in the major oxidative pathway of alcohol metabolism (Mello et al., 2008). Although several isozymes of ALDH have been identified, only cytosolic ALDH1 and mitochondrial ALDH2 are known to metabolize acetaldehyde (Moon et al., 2007). In the present study, *Adh6* and *Aldh1a7* were upregulated after 1 day of alcohol–CGX treatment, whereas *Adh4* and *Aldh1a1* were upregulated at 4 weeks of treatment (Fig. 3); however, consistent with the finding observed in the microarray analysis, *Aldh1b1* was downregulated at 4 weeks of treatment. Nevertheless, other members of these gene families were not significantly affected by alcohol–CGX treatment. Further molecular studies are needed to determine the cause–effect relationships among the genes altered by alcohol–CGX treatment.

VEGF is generally believed to drive fibrosis by promoting angiogenesis (Corpechot et al., 2002; Yoshiji et al., 2003); however, its role in fibrogenesis is not clear. VEGF transcription is regulated by TGF-β1 (Lee et al., 2008), and it has been reported that CGX treatment significantly reduced TGF-β in a TAA-induced chronic liver injury model (Kwak et al., 2011). In the present study, we assessed the expression levels of four growth factors involved in producing anti-fibrotic effects, including *Vegf*, by using RT-PCR. The microarray results indicated that only *Vegf* was significantly affected by alcohol–CGX treatment. However, RT-PCR evaluation indicated that the expression levels of these growth factors were decreased in a dose-dependent manner upon treatment with alcohol–CGX, when compared with those upon alcohol-only treatment (Fig. 4B and C). The VEGF protein level was determined by ELISA (Fig. 4B), and the results showed that at 4-week time point, the group treated with alcohol–CGX at 100 and 200 mg/kg/day exhibited slightly lower VEGF levels, when compared with those treated with alcohol only. This indicated that inhibition of VEGF through TGF-β1 by CGX in alcohol-induced rat livers plays an important role in the suppression of liver fibrosis.

In summary, the data obtained in the present study can facilitate understanding of the anti-fibrotic effects of the herbal formula CGX. The VEGF mRNA and protein expressions were found to increase during the development of alcohol-induced fibrosis, whereas alcohol–CGX treatment was noted to attenuate fibrogenesis through suppression of growth factors, including VEGF, and regulation of the *Adh* and *Aldh* gene families in the liver.

5. Conclusion

In the present study, we employed microarray technology to investigate the mechanisms underlying the anti-fibrotic effects of CGX in a rat model of alcoholic liver fibrosis. The results obtained indicate that suppression of the VEGF protein and regulation of the *Adh* and *Aldh* gene families by CGX have anti-fibrotic effects on chronic alcohol consumption status. Furthermore, these gene expression profiles also help us to better understand how multiple compounds such as CGX exert pharmaceutical effects on alcohol liver injury.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.jep.2013.09.035.

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